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# =====
# AIM:
# Read a captured video in Python and display
# the video in slow motion and fast motion.
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# Upload Video
from google.colab import files
uploaded = files.upload()

# Get uploaded video filename
video_file = list(uploaded.keys())[0]

import cv2
from IPython.display import HTML, display

# Open video
cap = cv2.VideoCapture(video_file)

# Check if video opened successfully
if not cap.isOpened():
    print("Error: Cannot open video.")
else:
    # Get video properties
    fps = cap.get(cv2.CAP_PROP_FPS)
    width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
    height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))

    print("Video Loaded Successfully")
    print("FPS:", fps)
    print("Resolution:", width, "x", height)

    # Define codec
    fourcc = cv2.VideoWriter_fourcc(*'mp4v')

    # Create output video writers
    slow_writer = cv2.VideoWriter(
        "slow_motion.mp4",
        fourcc,
        fps/2,
        (width, height)
    )

    fast_writer = cv2.VideoWriter(
        "fast_motion.mp4",
        fourcc,
        fps*2,
        (width, height)
    )

    # Read frames and write

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while True:
    ret, frame = cap.read()

    if not ret:
        break

    slow_writer.write(frame)
    fast_writer.write(frame)

# Release resources
cap.release()
slow_writer.release()
fast_writer.release()

print("Slow Motion and Fast Motion Videos Created Successfully!")

# Convert videos to browser-compatible format
!apt-get -qq install ffmpeg

!ffmpeg -loglevel error -y -i slow_motion.mp4 -vcodec libx264 slow_output.mp4
!ffmpeg -loglevel error -y -i fast_motion.mp4 -vcodec libx264 fast_output.mp4

print("Conversion Completed")

# Display Slow Motion Video
print("Slow Motion Video")
display(HTML("""
<video width="500" controls>
    <source src="slow_output.mp4" type="video/mp4">
</video>
"""))

# Display Fast Motion Video
print("Fast Motion Video")
display(HTML("""
<video width="500" controls>
    <source src="fast_output.mp4" type="video/mp4">
</video>
"""))

# Download the output videos
files.download("slow_output.mp4")
files.download("fast_output.mp4")
```

[Choose files](#) 11943452_...0fps (2).mp4

11943452_360_640_60fps (2).mp4(video/mp4) - 4603178 bytes, last modified: 27/07/2026 - 100% done

Saving 11943452_360_640_60fps (2).mp4 to 11943452_360_640_60fps (2).mp4

Video Loaded Successfully

FPS: 60.0

Resolution: 360 x 640

Slow Motion and Fast Motion Videos Created Successfully!

Conversion Completed

Slow Motion Video

0:00



Fast Motion Video

0:00

